Details on addressing incomplete soil sample data: Supporting document IndigoCarbon_US-1_2024_0063

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1 Executive summary

This document fulfills the requirements of Supporting Document IndigoCarbon_US-1_2023_0062 (2023). Based on the results presented below, we have concluded that there is minimal potential for bias due to missing data. Therefore, we are not post-stratifying and are only submitting an unadjusted analysis of reversible emissions reduction. While Supporting Document IndigoCarbon_US-1_2023_0062 (2023) also allows for the use of post-stratification to reduce sampling variance, we have chosen not to pursue this strategy.

Note that in the unadjusted analysis, missing data likely results in increased variance for estimated emissions reduction, and thus a larger uncertainty deduction. Consequently, the unadjusted analysis is automatically penalized for missing data by way of an increased uncertainty deduction. The present document is focused on the potential for missing data to cause bias, and whether any actions are required to account for potential concerns beyond the automatic increase in the uncertainty deduction.

2 Overview

As discussed in Supporting Document IndigoCarbon_US-1_2023_0062 (2023), incomplete (missing) soil sample data have the potential to bias estimates of emissions reduction. Supporting Document IndigoCarbon_US-1_2023_0062 (2023) lays out a series of steps for addressing incomplete soil sample data, and Figure 1 highlights those steps relevant to the current reporting period. The present document presents details for each step shown in Figure 1 that is applicable to the current reporting period.

As described in Supporting Document IndigoCarbon_US-1_2024_0068 (2024), the project has strata referred to as Analysis Strata A, B, C, D, E, F, G, and H. Reversible emissions reduction were quantified separately for each analysis stratum (with the exception that model variance, and thus the uncertainty deduction, was calculated across all analysis strata; see Supporting Document IndigoCarbon_US-1_2023_0062 (2023) Section 6.2.3 for details). Consequently, the assessments in this document are also specific to each analysis stratum.

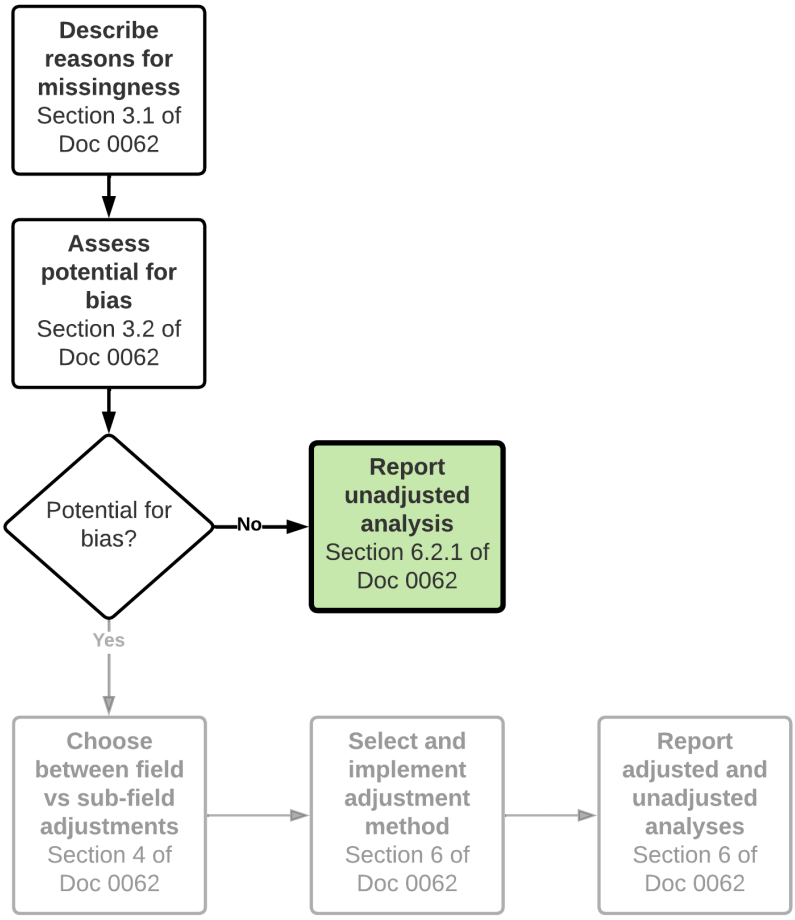


Figure 1: Overall steps and decisions in addressing incomplete soil samples as implemented for this reporting period. Section numbers refer to Supporting Document `IndigoCarbon_US-1_2023_0062` (2023). Grayed out boxes are not relevant for this reporting period; **the three bottom boxes related to adjusted analyses are out-of-scope for the current document.**

3 Reasons for missingness

Sample points that were selected by a sampling plan and intended to be used to quantify reversible emission reductions can be missing due to (1) soil sampling errors, i.e. unable to reliably measure initial percent soil organic carbon (%SOC), (2) biogeochemical model run failures, or (3) quality check failures after model runs. Table 1 tabulates the number of points that are missing from Indigo's analysis due to each of these reasons.

Table 1: Reasons why reversible emissions reduction could not be quantified at points selected by sampling plans

Analysis Stratum	Reason for missingness	Count (% of total planned sample points within analysis stratum)
A	Missing prior to model run (see Table 2)	1,137 (15.70% of 7242)
	Failed during model run	12 (0.17% of 7242)
	Did not pass quality check after model run*	0 (0.00% of 7242)
	All reasons	1,149 (15.87% of 7242)
B	Missing prior to model run (see Table 2)	4,640 (39.69% of 11692)
	Failed during model run	69 (0.59% of 11692)
	Did not pass quality check after model run*	0 (0.00% of 11692)
	All reasons	4,709 (40.28% of 11692)
C	Missing prior to model run (see Table 2)	1,970 (73.62% of 2676)
	Failed during model run	0 (0.00% of 2676)
	Did not pass quality check after model run*	0 (0.00% of 2676)
	All reasons	1,970 (73.62% of 2676)
D	Missing prior to model run (see Table 2)	756 (32.84% of 2302)
	Failed during model run	74 (3.21% of 2302)
	Did not pass quality check after model run*	0 (0.00% of 2302)
	All reasons	830 (36.06% of 2302)
E	Missing prior to model run (see Table 2)	1,725 (23.83% of 7240)
	Failed during model run	173 (2.39% of 7240)
	Did not pass quality check after model run*	0 (0.00% of 7240)
	All reasons	1,898 (26.22% of 7240)
F	Missing prior to model run (see Table 2)	3,256 (35.74% of 9111)
	Failed during model run	376 (4.13% of 9111)
	Did not pass quality check after model run*	0 (0.00% of 9111)
	All reasons	3,632 (39.86% of 9111)
G	Missing prior to model run (see Table 2)	255 (17.60% of 1449)
	Failed during model run	4 (0.28% of 1449)
	Did not pass quality check after model run*	0 (0.00% of 1449)
	All reasons	259 (17.87% of 1449)

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Table 1: Reasons why reversible emissions reduction could not be quantified at points selected by sampling plans

Analysis Stratum	Reason for missingness	Count (% of total planned sample points within analysis stratum)
H	Missing prior to model run (see Table 2)	2,994 (100.00% of 2994)
	Failed during model run	0 (0.00% of 2994)
	Did not pass quality check after model run*	0 (0.00% of 2994)
	All reasons	2,994 (100.00% of 2994)
Total	Missing prior to model run (see Table 2)	16,733 (37.43% of 44706)
	Failed during model run	708 (1.58% of 44706)
	Did not pass quality check after model run*	0 (0.00% of 44706)
	All reasons	17,441 (39.01% of 44706)

*Modeled changes in SOC for project and baseline scenarios, as well as emissions reduction, were checked to ensure they were less than 5,000 g/m^2 in magnitude. If modeled changes or emissions reduction were larger than 5,000 g/m^2 in magnitude at a given point, that point was discarded. See Supporting Document`IndigoCarbon_US-1_2023_0065` (2023) for details.

Table 2 provides more detailed tabulations on soil sampling errors (missingness that occurs prior to running the model), including missing %SOC, bulk density, and composite texture and pH samples.

Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
A	Call plan opened but field not yet sampled	0 (0%)	2 (0.09% of 2,360)	2 (0.30% of 663)	2 (0.30% of 663)
	Did not pass quality check prior to model run‡	915 (12.91% of 7,242)	34 (1.47% of 2,360)	3 (0.45% of 663)	3 (0.45% of 663)
	Field boundary expanded after randomization extsuperscript‡	153 (2.11% of 7,242)	51 (6.48% of 2,360)	0 (0%)	0 (0%)
	Point outside of boundary	2 (0.03% of 7,242)	0 (0%)	0 (0%)	0 (0%)
	Unknown	67 (0.95% of 7,242)	56 (2.43% of 2,360)	92 (13.88% of 663)	97 (14.63% of 663)
	All reasons	1,137 (15.99% of 7,242)	143 (10.47% of 2,360)	97 (14.63% of 663)	102 (15.38% of 663)
B	Call plan opened but field not yet sampled	116 (1.47% of 11,692)	39 (1.52% of 3,841)	23 (3.90% of 590)	23 (3.90% of 590)
	Did not pass quality check prior to model run‡	643 (8.16% of 11,692)	123 (4.78% of 3,841)	13 (2.20% of 590)	13 (2.20% of 590)
	Field boundary expanded after randomization extsuperscript‡	168 (2.09% of 11,692)	56 (6.39% of 3,841)	0 (0%)	0 (0%)
	Hard soil	1 (0.01% of 11,692)	1 (0.04% of 3,841)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
	Location inaccessible	7 (0.09% of 11,692)	3 (0.12% of 3,841)	2 (0.34% of 590)	2 (0.34% of 590)
	Point in water	4 (0.05% of 11,692)	1 (0.04% of 3,841)	0 (0%)	0 (0%)
	Point outside of boundary	11 (0.14% of 11,692)	4 (0.16% of 3,841)	2 (0.34% of 590)	2 (0.34% of 590)
	Sampling point not assigned to field staff	3,640 (31.13% of 11,692)	1,213 (94.77% of 3,841)	0 (0%)	0 (0%)
	Unable to sample to full depth	25 (0.32% of 11,692)	5 (0.19% of 3,841)	5 (0.85% of 590)	5 (0.85% of 590)
	Unknown	25 (0.32% of 11,692)	17 (0.66% of 3,841)	77 (13.05% of 590)	79 (13.39% of 590)
	All reasons	4,640 (43.77% of 11,692)	1,462 (108.66% of 3,841)	122 (20.68% of 590)	124 (21.02% of 590)
	Call plan opened but field not yet sampled	222 (22.58% of 2,676)	76 (23.03% of 894)	10 (16.13% of 62)	10 (16.13% of 62)
	Did not pass quality check prior to model run‡	48 (4.88% of 2,676)	9 (2.73% of 894)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
D	Field boundary expanded after randomization extsuperscript‡	1 (0.10% of 2,676)	0 (0.30% of 894)	0 (0%)	0 (0%)
	Location inaccessible	1 (0.10% of 2,676)	0 (0%)	0 (0%)	0 (0%)
	Point outside of boundary	3 (0.31% of 2,676)	2 (0.61% of 894)	2 (3.23% of 62)	2 (3.23% of 62)
	Sampling point not assigned to field staff	1,692 (63.23% of 2,676)	564 (189.26% of 894)	0 (0%)	0 (0%)
	Unable to sample to full depth	1 (0.10% of 2,676)	0 (0%)	0 (0%)	0 (0%)
	Unknown	2 (0.20% of 2,676)	0 (0%)	6 (9.68% of 62)	6 (9.68% of 62)
	All reasons	1,970 (91.51% of 2,676)	651 (215.93% of 894)	18 (29.03% of 62)	18 (29.03% of 62)
	Call plan opened but field not yet sampled	24 (1.33% of 2,302)	8 (1.32% of 768)	1 (0.88% of 113)	1 (0.88% of 113)
	Did not pass quality check prior to model run‡	224 (12.39% of 2,302)	21 (3.48% of 768)	4 (3.54% of 113)	4 (3.54% of 113)
	Field boundary expanded after randomization extsuperscript‡	4 (0.22% of 2,302)	1 (0.66% of 768)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
E	Point outside of boundary	2 (0.11% of 2,302)	1 (0.17% of 768)	0 (0%)	0 (0%)
	Sampling point not assigned to field staff	490 (21.29% of 2,302)	163 (63.80% of 768)	0 (0%)	0 (0%)
	Seasonal delay	0 (0%)	0 (0%)	1 (0.88% of 113)	1 (0.88% of 113)
	Unknown	12 (0.66% of 2,302)	4 (0.66% of 768)	19 (16.81% of 113)	19 (16.81% of 113)
	All reasons	756 (36.00% of 2,302)	198 (70.09% of 768)	25 (22.12% of 113)	25 (22.12% of 113)
	Call plan opened but field not yet sampled	198 (3.28% of 7,240)	67 (3.31% of 2,426)	14 (3.14% of 446)	14 (3.14% of 446)
	Did not pass quality check prior to model run‡	271 (4.49% of 7,240)	109 (5.38% of 2,426)	16 (3.59% of 446)	16 (3.59% of 446)
	Field boundary expanded after randomization extsuperscript‡	38 (0.63% of 7,240)	13 (1.86% of 2,426)	0 (0%)	0 (0%)
	Hard soil	4 (0.07% of 7,240)	0 (0%)	0 (0%)	0 (0%)
	Location inaccessible	9 (0.15% of 7,240)	4 (0.20% of 2,426)	2 (0.45% of 446)	2 (0.45% of 446)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
F	Point in water	1 (0.02% of 7,240)	0 (0%)	0 (0%)	0 (0%)
	Point outside of boundary	9 (0.15% of 7,240)	6 (0.30% of 2,426)	3 (0.67% of 446)	3 (0.67% of 446)
	Sampling point not assigned to field staff	1,164 (16.08% of 7,240)	388 (47.98% of 2,426)	0 (0%)	0 (0%)
	Unable to sample to full depth	5 (0.08% of 7,240)	2 (0.10% of 2,426)	2 (0.45% of 446)	2 (0.45% of 446)
	Unknown	26 (0.43% of 7,240)	11 (0.54% of 2,426)	70 (15.70% of 446)	70 (15.70% of 446)
	All reasons	1,725 (25.36% of 7,240)	600 (59.67% of 2,426)	107 (23.99% of 446)	107 (23.99% of 446)
	Call plan opened but field not yet sampled	1,494 (20.05% of 9,111)	499 (20.05% of 3,041)	119 (20.14% of 591)	119 (20.14% of 591)
	Did not pass quality check prior to model run‡	47 (0.63% of 9,111)	87 (3.50% of 3,041)	18 (3.05% of 591)	17 (2.88% of 591)
	Field boundary expanded after randomization extsuperscript‡	1 (0.01% of 9,111)	0 (0.04% of 3,041)	0 (0%)	0 (0%)
	Hard soil	3 (0.04% of 9,111)	1 (0.04% of 3,041)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
G	Location inaccessible	5 (0.07% of 9,111)	4 (0.16% of 3,041)	1 (0.17% of 591)	1 (0.17% of 591)
	Point in water	11 (0.15% of 9,111)	5 (0.20% of 3,041)	2 (0.34% of 591)	2 (0.34% of 591)
	Point outside of boundary	22 (0.30% of 9,111)	9 (0.36% of 3,041)	3 (0.51% of 591)	3 (0.51% of 591)
	Sampling point not assigned to field staff	1,657 (18.19% of 9,111)	552 (54.49% of 3,041)	0 (0%)	0 (0%)
	Unknown	16 (0.21% of 9,111)	6 (0.24% of 3,041)	30 (5.08% of 591)	47 (7.95% of 591)
	All reasons	3,256 (39.64% of 9,111)	1,163 (79.08% of 3,041)	173 (29.27% of 591)	189 (31.98% of 591)
	Call plan opened but field not yet sampled	206 (14.22% of 1,449)	68 (14.11% of 482)	18 (15.25% of 118)	18 (15.25% of 118)
	Did not pass quality check prior to model run[‡]	17 (1.17% of 1,449)	3 (0.62% of 482)	0 (0%)	0 (0%)
	Hard soil	3 (0.21% of 1,449)	1 (0.21% of 482)	1 (0.85% of 118)	1 (0.85% of 118)
	Location inaccessible	5 (0.35% of 1,449)	1 (0.21% of 482)	1 (0.85% of 118)	1 (0.85% of 118)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
H	Point outside of boundary	9 (0.62% of 1,449)	0 (0%)	0 (0%)	0 (0%)
	Unknown	15 (1.04% of 1,449)	5 (1.04% of 482)	7 (5.93% of 118)	7 (5.93% of 118)
	All reasons	255 (17.60% of 1,449)	78 (16.18% of 482)	27 (22.88% of 118)	27 (22.88% of 118)
	Call plan opened but field not yet sampled	2,994 (100.00% of 2,994)	998 (100.00% of 998)	246 (100.00% of 246)	246 (100.00% of 246)
	All reasons	2,994 (100.00% of 2,994)	998 (100.00% of 998)	246 (100.00% of 246)	246 (100.00% of 246)

Notes for Table 2:

† A sample could fail the quality check prior to the biogeochemical model run if any of the following occurred:

- The sample QR code was not valid, the sample QR code was not unique, or the sample was soft-deleted[1] in the database (because it failed sample QC)
- The laboratory QR code was not valid, the laboratory QR code was not unique, or the laboratory result was soft-deleted (because it failed laboratory QC)
- The sample was not taken as prescribed
- An incorrect analysis was done
- The shipment was delayed. A shipment is classified as delayed if: (1) there is more than 21 days between when the sample was collected and received by the lab (used if a lab received date is available), or (2) there is more than 49 days between when the sample was collected and when its result was returned from the lab (used if a lab received date is not available)
- The sample depth was not adequate
- The sampling point was moved beyond the standard operating procedure (SOP) limit (see `IndigoCarbon_US-1_2023_0001`, 2024 for details). Note that points missing due to "Protocol not followed (sample collected at wrong location)" covers all reasons related to taking a sample at the wrong location other than moving the point beyond the SOP limit.

‡ In rare cases, a field boundary was changed after randomization (see Supporting Document `IndigoCarbon_US-1_2024_0034` (2023)). We generated random points in the newly added land at the same sampling density (number of planned points per acre) as in the rest of the pre-stratum (note: not analysis stratum but pre-stratum, i.e., the original stratum at the time of randomization; see `IndigoCarbon_US-1_2024_0068`, 2024 for details on that distinction). Consequently, we only generated points on newly added land if the increase in area multiplied by the sampling density was greater than 1, i.e. if we would have expected sampling points to have been randomly selected in the newly added land had the land been in the pre-stratum at the time of randomization.

4 Assessment of potential bias

Tables 3–10 compare the attributes of i) all points in the sample versus ii) points at which emissions reduction were successfully modeled. These comparisons follow Section 3.2 of Supporting Document `IndigoCarbon_US-1_2023_0062` (2023). For categorical attributes, counts and percents are shown in the tables (for each attribute, percents sum to 1 within each column). For continuous attributes, the mean and standard deviation are shown. The right-most column shows the standardized mean difference (SMD) (see Equations (1), (2), and (58) in `IndigoCarbon_US-1_2023_0062` (2023)).

In Tables 3–10, the total number of points in the sample are shown under "All", and the total number of observed points is shown under "Observed". Some attributes, such as weather and management practices, change over time. To summarize the spatial-temporal data into summaries for

[1] When a sample is "soft-deleted," it is flagged as unusable, but is not actually erased or removed from the database.

each point, we took the temporal mean for continuous variables (averaged over the project period). For binary variables, such as whether a certain practice change was adopted, we summarized the attribute as "True" if it occurred at any time during the project period, and "False" otherwise.

We took the following approach to collect field attribute data at each point:

- Points for which (latitude, longitude) coordinates were generated (some have not been generated yet because we wait to generate them until visiting the field): Attributes are pulled based on the geographic location of the planned soil samples.
- Points for which (latitude, longitude) coordinates have not yet been generated: We randomly generated (latitude, longitude) coordinates in affected fields so that within each field the total number of generated points is equal to the number of points that will be generated by the call plan when the sampling crews are eventually instructed to sample the field.[2] Attributes were then pulled based on the generated coordinates.
- Additional points that might have been generated by Indigo's call plan in land that was added after randomization (see ‡ note above): We randomly generated (latitude, longitude) coordinates in newly added land so that the number of points per acre would be the same as that observed in the rest of the pre-stratum. Attributes were then pulled based on the generated coordinates.

We also note that for attributes with more than two categories, the SMD cannot be calculated if there are two or more categories with zero counts for the number of observed samples. If applicable, these SMDs appear as NaN in Tables 3–10 and will not be shown in Figures 2–9.

Please see Appendix A for the definitions of attributes in Tables 3–10, and please see Table 12 for the definitions of crop functional groups.

[2]The actual software for soil sampling may generate different coordinates.

Table 3: Pre-Stratum A

		All (7,242)	Observed (6,093)	SMD
bulk_density_g_per_cm3		1.36 (0.11)	1.36 (0.12)	0.021
average_tmax_c		19.54 (4.46)	19.44 (4.56)	-0.022
clay_pct		25.08 (9.75)	25.03 (9.95)	-0.005
sand_pct		22.02 (18.88)	22.70 (19.52)	0.036
elevation_m		435.40 (260.79)	414.84 (235.93)	-0.079
silt_pct		52.78 (17.03)	52.13 (17.49)	-0.038
average_precip_cm		0.20 (0.06)	0.20 (0.06)	0.043
percentage_residue_removed		0.68 (7.00)	0.68 (7.01)	-0.001
tillage_depth_cm		2.58 (4.11)	2.81 (4.23)	0.054
average_srad_mj_m2		0.68 (0.07)	0.67 (0.07)	-0.031
soil_ph		6.75 (0.79)	6.72 (0.79)	-0.038
average_relative_humidity_perc		63.82 (8.73)	64.31 (8.34)	0.056
soil_carbon_pct		1.59 (1.86)	1.63 (2.00)	0.021
continuous_deep_till_to_no_till	False	7,242 (100.00%)	6,093 (100%)	< 0.001
cover_crop_introduction	False	6,179 (85.32%)	5,155 (84.6%)	0.02
	True	1,063 (14.68%)	938 (15.4%)	
fertilizer_application_delayed	False	6,603 (91.18%)	5,668 (93.0%)	0.065
	True	639 (8.82%)	425 (7.0%)	
fertilizer_method_impact_reduction	False	7,120 (98.32%)	5,982 (98.2%)	0.011
	True	122 (1.68%)	111 (1.8%)	
legume_introduced_in_cover_crop_mix	False	7,192 (99.31%)	6,057 (99.4%)	0.012
	True	50 (0.69%)	36 (0.6%)	
longer_duration_cover_crop	False	7,001 (96.67%)	5,868 (96.3%)	0.02
	True	241 (3.33%)	225 (3.7%)	
new_cash_crops_in_rotation	False	5,894 (81.39%)	4,996 (82.0%)	0.016
	True	1,348 (18.61%)	1,097 (18.0%)	
nitrogen_stabilizer_added	False	7,063 (97.53%)	5,991 (98.3%)	0.051
	True	179 (2.47%)	102 (1.7%)	

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Table 3: Pre-Stratum A

		All (7,242)	Observed (6,093)	SMD
no_till_introduced	True	3,990 (55.10%)	3,162 (51.9%)	0.064
	False	3,252 (44.90%)	2,931 (48.1%)	
tillage_impact_reduction	True	4,945 (68.28%)	4,038 (66.3%)	0.043
	False	2,297 (31.72%)	2,055 (33.7%)	
tillage_passes_reduction	True	4,366 (60.29%)	3,512 (57.6%)	0.054
	False	2,876 (39.71%)	2,581 (42.4%)	
corn	False	4,040 (55.79%)	3,467 (56.9%)	0.022
	True	3,202 (44.21%)	2,626 (43.1%)	
cotton	False	6,761 (93.36%)	5,641 (92.6%)	0.031
	True	481 (6.64%)	452 (7.4%)	
soybean	False	4,043 (55.83%)	3,370 (55.3%)	0.01
	True	3,199 (44.17%)	2,723 (44.7%)	
wheat	False	4,621 (63.81%)	3,814 (62.6%)	0.025
	True	2,621 (36.19%)	2,279 (37.4%)	
fallow	False	6,407 (88.47%)	5,345 (87.7%)	0.023
	True	835 (11.53%)	748 (12.3%)	
chisel	False	7,075 (97.69%)	5,936 (97.4%)	0.018
	True	167 (2.31%)	157 (2.6%)	
deep_ripper	False	7,242 (100.00%)	6,093 (100%)	< 0.001
disk	False	6,728 (92.90%)	5,618 (92.2%)	0.027
	True	514 (7.10%)	475 (7.8%)	
finisher	False	7,057 (97.45%)	5,920 (97.2%)	0.018
	True	185 (2.55%)	173 (2.8%)	
moldboard	False	7,242 (100.00%)	6,093 (100%)	< 0.001
ridge_till	False	7,242 (100.00%)	6,093 (100%)	< 0.001
row_cultivator	False	7,234 (99.89%)	6,085 (99.9%)	0.006
	True	8 (0.11%)	8 (0.1%)	
strip_till	False	6,262 (86.47%)	5,172 (84.9%)	0.046

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Table 3: Pre-Stratum A

		All (7,242)	Observed (6,093)	SMD
vertical_till	True	980 (13.53%)	921 (15.1%)	
	False	6,428 (88.76%)	5,407 (88.7%)	0.001
	True	814 (11.24%)	686 (11.3%)	
lrr_code	M	3,922 (54.16%)	3,531 (58.0%)	0.101
	H	1,752 (24.19%)	1,322 (21.7%)	
	F	1,197 (16.53%)	989 (16.2%)	
	P	207 (2.86%)	108 (1.8%)	
	I	64 (0.88%)	43 (0.7%)	
	R	41 (0.57%)	41 (0.7%)	
	J	36 (0.50%)	36 (0.6%)	
	K	12 (0.17%)	12 (0.2%)	
	T	11 (0.15%)	11 (0.2%)	

Table 4: Pre-Stratum B

	All (11,692)	Observed (6,983)	SMD
bulk_density_g_per_cm3	1.38 (0.12)	1.38 (0.10)	-0.048
average_tmax_c	18.85 (3.67)	18.63 (3.50)	-0.06
clay_pct	25.31 (8.63)	26.00 (7.78)	0.08
sand_pct	22.90 (20.33)	20.81 (17.63)	-0.102
elevation_m	366.40 (177.34)	351.05 (170.85)	-0.087
silt_pct	51.69 (16.54)	53.18 (15.04)	0.091
average_precip_cm	0.22 (0.07)	0.22 (0.07)	0.026
percentage_residue_removed	0.74 (7.67)	1.02 (9.08)	0.036
tillage_depth_cm	2.21 (3.82)	2.33 (3.95)	0.033
average_srad_mj_m2	0.67 (0.05)	0.66 (0.04)	-0.072

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Table 4: Pre-Stratum B

		All (11,692)	Observed (6,983)	SMD
soil_ph		6.54 (0.68)	6.55 (0.62)	0.025
average_relative_humidity_perc		66.42 (6.29)	67.22 (5.87)	0.128
soil_carbon_pct		1.66 (2.32)	1.70 (2.16)	0.019
continuous_deep_till_to_no_till	False	11,656 (99.69%)	6,961 (99.7%)	0.001
	True	36 (0.31%)	22 (0.3%)	
cover_crop_introduction	False	8,939 (76.45%)	5,546 (79.4%)	0.07
	True	2,753 (23.55%)	1,437 (20.6%)	
fertilizer_application_delayed	False	10,808 (92.44%)	6,667 (95.5%)	0.115
	True	884 (7.56%)	316 (4.5%)	
fertilizer_method_impact_reduction	False	11,134 (95.23%)	6,708 (96.1%)	0.039
	True	558 (4.77%)	275 (3.9%)	
legume_introduced_in_cover_crop_mix	False	11,572 (98.97%)	6,918 (99.1%)	0.009
	True	120 (1.03%)	65 (0.9%)	
longer_duration_cover_crop	False	10,798 (92.35%)	6,510 (93.2%)	0.033
	True	894 (7.65%)	473 (6.8%)	
new_cash_crops_in_rotation	False	9,767 (83.54%)	6,002 (86.0%)	0.065
	True	1,925 (16.46%)	981 (14.0%)	
nitrogen_stabilizer_added	False	11,579 (99.03%)	6,942 (99.4%)	0.039
	True	113 (0.97%)	41 (0.6%)	
no_till_introduced	True	6,771 (57.91%)	4,100 (58.7%)	0.016
	False	4,921 (42.09%)	2,883 (41.3%)	
tillage_impact_reduction	True	7,903 (67.59%)	4,819 (69.0%)	0.03
	False	3,789 (32.41%)	2,164 (31.0%)	
tillage_passes_reduction	True	7,595 (64.96%)	4,583 (65.6%)	0.014
	False	4,097 (35.04%)	2,400 (34.4%)	
corn	True	5,928 (50.70%)	3,216 (46.1%)	0.093
	False	5,764 (49.30%)	3,767 (53.9%)	
cotton	False	11,482 (98.20%)	6,896 (98.8%)	0.041

Continued on next page

Table 4: Pre-Stratum B

		All (11,692)	Observed (6,983)	SMD
	True	210 (1.80%)	87 (1.2%)	
soybean	True	6,317 (54.03%)	3,584 (51.3%)	0.054
	False	5,375 (45.97%)	3,399 (48.7%)	
wheat	True	5,965 (51.02%)	3,125 (44.8%)	0.125
	False	5,727 (48.98%)	3,858 (55.2%)	
fallow	False	11,193 (95.73%)	6,695 (95.9%)	0.007
	True	499 (4.27%)	288 (4.1%)	
chisel	False	11,253 (96.25%)	6,628 (94.9%)	0.07
	True	439 (3.75%)	355 (5.1%)	
deep_ripper	False	11,344 (97.02%)	6,719 (96.2%)	0.047
	True	348 (2.98%)	264 (3.8%)	
disk	False	10,954 (93.69%)	6,622 (94.8%)	0.047
	True	738 (6.31%)	361 (5.2%)	
finisher	False	10,357 (88.58%)	6,178 (88.5%)	0.003
	True	1,335 (11.42%)	805 (11.5%)	
moldboard	False	11,692 (100.00%)	6,983 (100%)	< 0.001
ridge_till	False	11,692 (100.00%)	6,983 (100%)	< 0.001
row_cultivator	False	11,604 (99.25%)	6,909 (98.9%)	0.036
	True	88 (0.75%)	74 (1.1%)	
strip_till	False	10,999 (94.07%)	6,587 (94.3%)	0.011
	True	693 (5.93%)	396 (5.7%)	
vertical_till	False	10,592 (90.59%)	6,300 (90.2%)	0.013
	True	1,100 (9.41%)	683 (9.8%)	
lrr_code	M	5,842 (49.97%)	4,093 (58.6%)	0.207
	H	2,645 (22.62%)	1,156 (16.6%)	
	F	1,865 (15.95%)	1,010 (14.5%)	
	P	420 (3.59%)	155 (2.2%)	
	L	322 (2.75%)	212 (3.0%)	

Continued on next page

Table 4: Pre-Stratum B

		All (11,692)	Observed (6,983)	SMD
	O	202 (1.73%)	129 (1.8%)	
	N	175 (1.50%)	62 (0.9%)	
	R	103 (0.88%)	66 (0.9%)	
	K	87 (0.74%)	69 (1.0%)	
	G	19 (0.16%)	19 (0.3%)	
	T	12 (0.10%)	12 (0.2%)	

Table 5: Pre-Stratum C

		All (2,676)	Observed (706)	SMD
bulk_density_g_per_cm3		1.36 (0.07)	1.37 (0.07)	0.15
average_tmax_c		19.65 (2.59)	18.21 (1.78)	-0.554
clay_pct		28.12 (10.71)	27.40 (7.75)	-0.067
sand_pct		18.74 (17.10)	22.44 (14.25)	0.217
elevation_m		342.79 (226.72)	315.10 (137.41)	-0.122
silt_pct		53.15 (14.77)	50.16 (13.08)	-0.202
average_precip_cm		0.24 (0.06)	0.21 (0.04)	-0.494
percentage_residue_removed		0.00 (0.00)	0.00 (0.00)	NaN
tillage_depth_cm		2.06 (4.14)	4.13 (5.09)	0.498
average_srad_mj_m2		0.67 (0.05)	0.65 (0.02)	-0.47
soil_ph		6.59 (0.72)	6.52 (0.54)	-0.108
average_relative_humidity_perc		66.57 (7.12)	68.10 (3.17)	0.215
soil_carbon_pct		1.50 (0.69)	1.78 (0.67)	0.416
continuous_deep_till_to_no_till	False	2,676 (100.00%)	706 (100%)	< 0.001
cover_crop_introduction	True	1,574 (58.82%)	121 (17.1%)	0.847
	False	1,102 (41.18%)	585 (82.9%)	

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Table 5: Pre-Stratum C

		All (2,676)	Observed (706)	SMD
fertilizer_application_delayed	False	1,508 (56.35%)	642 (90.9%)	0.697
	True	1,168 (43.65%)	64 (9.1%)	
fertilizer_method_impact_reduction	False	2,591 (96.82%)	682 (96.6%)	0.013
	True	85 (3.18%)	24 (3.4%)	
legume_introduced_in_cover_crop_mix	False	2,676 (100.00%)	706 (100%)	< 0.001
longer_duration_cover_crop	False	2,664 (99.55%)	694 (98.3%)	0.187
	True	12 (0.45%)	12 (1.7%)	
new_cash_crops_in_rotation	False	2,476 (92.53%)	702 (99.4%)	0.263
	True	200 (7.47%)	4 (0.6%)	
nitrogen_stabilizer_added	False	2,676 (100.00%)	706 (100%)	< 0.001
no_till_introduced	True	1,789 (66.85%)	194 (27.5%)	0.836
	False	887 (33.15%)	512 (72.5%)	
tillage_impact_reduction	True	2,035 (76.05%)	294 (41.6%)	0.806
	False	641 (23.95%)	412 (58.4%)	
tillage_passes_reduction	True	2,005 (74.93%)	252 (35.7%)	0.905
	False	671 (25.07%)	454 (64.3%)	
corn	True	1,947 (72.76%)	390 (55.2%)	0.393
	False	729 (27.24%)	316 (44.8%)	
cotton	False	2,616 (97.76%)	706 (100%)	0.151
	True	60 (2.24%)	0 (0%)	
soybean	True	1,802 (67.34%)	417 (59.1%)	0.176
	False	874 (32.66%)	289 (40.9%)	
wheat	True	1,812 (67.71%)	163 (23.1%)	0.954
	False	864 (32.29%)	543 (76.9%)	
fallow	False	2,670 (99.78%)	706 (100%)	0.047
	True	6 (0.22%)	0 (0%)	
chisel	False	2,628 (98.21%)	670 (94.9%)	0.249
	True	48 (1.79%)	36 (5.1%)	

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Table 5: Pre-Stratum C

		All (2,676)	Observed (706)	SMD
deep_ripper	False	2,676 (100.00%)	706 (100%)	< 0.001
disk	False	2,472 (92.38%)	624 (88.4%)	0.15
	True	204 (7.62%)	82 (11.6%)	
finisher	False	2,496 (93.27%)	590 (83.6%)	0.387
	True	180 (6.73%)	116 (16.4%)	
moldboard	False	2,640 (98.65%)	671 (95.0%)	0.314
	True	36 (1.35%)	35 (5.0%)	
ridge_till	False	2,676 (100.00%)	706 (100%)	< 0.001
row_cultivator	False	2,664 (99.55%)	695 (98.4%)	0.166
	True	12 (0.45%)	11 (1.6%)	
strip_till	False	2,423 (90.55%)	552 (78.2%)	0.422
	True	253 (9.45%)	154 (21.8%)	
vertical_till	False	2,507 (93.68%)	619 (87.7%)	0.247
	True	169 (6.32%)	87 (12.3%)	
lrr_code	M	1,561 (58.33%)	373 (52.8%)	NaN
	H	431 (16.11%)	0 (0%)	
	L	312 (11.66%)	202 (28.6%)	
	N	252 (9.42%)	47 (6.7%)	
	F	72 (2.69%)	72 (10.2%)	
	J	36 (1.35%)	0 (0%)	
	P	12 (0.45%)	12 (1.7%)	

Table 6: Pre-Stratum D

		All (2,302)	Observed (1,472)	SMD
bulk_density_g_per_cm3		1.39 (0.11)	1.40 (0.08)	0.024
average_tmax_c		19.75 (3.83)	18.61 (3.22)	-0.299
clay_pct		26.78 (12.81)	25.45 (9.74)	-0.104
sand_pct		20.27 (17.57)	21.55 (18.18)	0.073
elevation_m		275.24 (198.81)	317.50 (188.13)	0.213
silt_pct		52.73 (16.90)	53.00 (16.57)	0.016
average_precip_cm		0.26 (0.08)	0.24 (0.05)	-0.326
percentage_residue_removed		0.33 (2.68)	0.52 (3.34)	0.07
tillage_depth_cm		4.18 (4.23)	3.37 (3.90)	-0.192
average_srad_mj_m2		0.68 (0.05)	0.66 (0.05)	-0.276
soil_ph		6.33 (0.65)	6.39 (0.60)	0.092
average_relative_humidity_perc		69.43 (6.22)	68.87 (6.44)	-0.09
soil_carbon_pct		1.66 (1.95)	1.71 (0.95)	0.023
continuous_deep_till_to_no_till	False	2,302 (100.00%)	1,472 (100%)	< 0.001
cover_crop_introduction	False	1,888 (82.02%)	1,163 (79.0%)	0.078
	True	414 (17.98%)	309 (21.0%)	
fertilizer_application_delayed	False	2,218 (96.35%)	1,449 (98.4%)	0.111
	True	84 (3.65%)	23 (1.6%)	
fertilizer_method_impact_reduction	False	2,300 (99.91%)	1,470 (99.9%)	0.017
	True	2 (0.09%)	2 (0.1%)	
legume_introduced_in_cover_crop_mix	False	2,218 (96.35%)	1,438 (97.7%)	0.071
	True	84 (3.65%)	34 (2.3%)	
longer_duration_cover_crop	False	2,182 (94.79%)	1,421 (96.5%)	0.079
	True	120 (5.21%)	51 (3.5%)	
new_cash_crops_in_rotation	False	1,983 (86.14%)	1,345 (91.4%)	0.151
	True	319 (13.86%)	127 (8.6%)	
nitrogen_stabilizer_added	False	2,302 (100.00%)	1,472 (100%)	< 0.001
no_till_introduced	False	1,451 (63.03%)	848 (57.6%)	0.112

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Table 6: Pre-Stratum D

		All (2,302)	Observed (1,472)	SMD
	True	851 (36.97%)	624 (42.4%)	
tillage_impact_reduction	True	1,675 (72.76%)	996 (67.7%)	0.115
	False	627 (27.24%)	476 (32.3%)	
tillage_passes_reduction	True	1,447 (62.86%)	844 (57.3%)	0.114
	False	855 (37.14%)	628 (42.7%)	
corn	True	1,370 (59.51%)	768 (52.2%)	0.15
	False	932 (40.49%)	704 (47.8%)	
cotton	False	2,242 (97.39%)	1,472 (100%)	0.164
	True	60 (2.61%)	0 (0%)	
soybean	False	1,188 (51.61%)	868 (59.0%)	0.147
	True	1,114 (48.39%)	604 (41.0%)	
wheat	False	1,516 (65.86%)	932 (63.3%)	0.054
	True	786 (34.14%)	540 (36.7%)	
fallow	False	2,234 (97.05%)	1,440 (97.8%)	0.046
	True	68 (2.95%)	32 (2.2%)	
chisel	False	2,088 (90.70%)	1,304 (88.6%)	0.073
	True	214 (9.30%)	168 (11.4%)	
deep_ripper	False	2,206 (95.83%)	1,460 (99.2%)	0.168
	True	96 (4.17%)	12 (0.8%)	
disk	False	1,954 (84.88%)	1,422 (96.6%)	0.327
	True	348 (15.12%)	50 (3.4%)	
finisher	False	1,882 (81.75%)	1,313 (89.2%)	0.193
	True	420 (18.25%)	159 (10.8%)	
moldboard	False	2,266 (98.44%)	1,472 (100%)	0.126
	True	36 (1.56%)	0 (0%)	
ridge_till	False	2,302 (100.00%)	1,472 (100%)	< 0.001
row_cultivator	False	1,966 (85.40%)	1,446 (98.2%)	0.363
	True	336 (14.60%)	26 (1.8%)	

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Table 6: Pre-Stratum D

		All (2,302)	Observed (1,472)	SMD
strip_till	False	2,016 (87.58%)	1,335 (90.7%)	0.094
	True	286 (12.42%)	137 (9.3%)	
vertical_till	False	1,886 (81.93%)	1,125 (76.4%)	0.143
	True	416 (18.07%)	347 (23.6%)	
lrr_code	M	1,397 (60.69%)	1,054 (71.6%)	0.403
	P	312 (13.55%)	24 (1.6%)	
	H	236 (10.25%)	180 (12.2%)	
	O	108 (4.69%)	31 (2.1%)	
	L	108 (4.69%)	48 (3.3%)	
	K	94 (4.08%)	88 (6.0%)	
	J	24 (1.04%)	24 (1.6%)	
	F	12 (0.52%)	12 (0.8%)	
	N	11 (0.48%)	11 (0.7%)	

Table 7: Pre-Stratum E

	All (7,240)	Observed (5,342)	SMD
bulk_density_g_per_cm3	1.35 (0.10)	1.34 (0.10)	-0.039
average_tmax_c	19.58 (3.07)	19.59 (3.04)	0.004
clay_pct	27.00 (8.52)	26.94 (7.55)	-0.007
sand_pct	16.70 (16.18)	16.72 (15.64)	0.001
elevation_m	317.77 (162.01)	321.29 (146.64)	0.022
silt_pct	56.30 (15.05)	56.35 (14.43)	0.003
average_precip_cm	0.24 (0.06)	0.23 (0.05)	-0.05
percentage_residue_removed	0.58 (5.82)	0.34 (4.52)	-0.041
tillage_depth_cm	3.82 (4.32)	3.86 (4.58)	0.01

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Table 7: Pre-Stratum E

		All (7,240)	Observed (5,342)	SMD
average_srad_mj_m2		0.67 (0.04)	0.67 (0.04)	0.001
soil_ph		6.32 (0.60)	6.35 (0.60)	0.047
average_relative_humidity_perc		66.73 (4.96)	66.57 (4.74)	-0.033
soil_carbon_pct		1.55 (0.81)	1.59 (0.82)	0.055
continuous_deep_till_to_no_till	False	7,228 (99.83%)	5,342 (100%)	0.041
	True	12 (0.17%)	0 (0%)	
cover_crop_introduction	False	6,009 (83.00%)	4,565 (85.5%)	0.065
	True	1,231 (17.00%)	777 (14.5%)	
fertilizer_application_delayed	False	6,901 (95.32%)	5,217 (97.7%)	0.111
	True	339 (4.68%)	125 (2.3%)	
fertilizer_method_impact_reduction	False	7,214 (99.64%)	5,318 (99.6%)	0.015
	True	26 (0.36%)	24 (0.4%)	
legume_introduced_in_cover_crop_mix	False	7,159 (98.88%)	5,275 (98.7%)	0.013
	True	81 (1.12%)	67 (1.3%)	
longer_duration_cover_crop	False	6,912 (95.47%)	5,054 (94.6%)	0.041
	True	328 (4.53%)	288 (5.4%)	
new_cash_crops_in_rotation	False	6,608 (91.27%)	4,935 (92.4%)	0.039
	True	632 (8.73%)	407 (7.6%)	
nitrogen_stabilizer_added	False	7,168 (99.01%)	5,314 (99.5%)	0.047
	True	72 (0.99%)	28 (0.5%)	
no_till_introduced	False	4,128 (57.02%)	2,961 (55.4%)	0.032
	True	3,112 (42.98%)	2,381 (44.6%)	
tillage_impact_reduction	True	5,527 (76.34%)	4,001 (74.9%)	0.034
	False	1,713 (23.66%)	1,341 (25.1%)	
tillage_passes_reduction	True	4,883 (67.44%)	3,646 (68.3%)	0.017
	False	2,357 (32.56%)	1,696 (31.7%)	
corn	True	3,650 (50.41%)	2,464 (46.1%)	0.086
	False	3,590 (49.59%)	2,878 (53.9%)	

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Table 7: Pre-Stratum E

		All (7,240)	Observed (5,342)	SMD
cotton	False	7,030 (97.10%)	5,141 (96.2%)	0.051
	True	210 (2.90%)	201 (3.8%)	
soybean	True	4,033 (55.70%)	2,742 (51.3%)	0.088
	False	3,207 (44.30%)	2,600 (48.7%)	
wheat	False	4,766 (65.83%)	3,493 (65.4%)	0.009
	True	2,474 (34.17%)	1,849 (34.6%)	
fallow	False	6,923 (95.62%)	5,115 (95.8%)	0.006
	True	317 (4.38%)	227 (4.2%)	
chisel	False	7,216 (99.67%)	5,330 (99.8%)	0.019
	True	24 (0.33%)	12 (0.2%)	
deep_ripper	False	7,152 (98.78%)	5,283 (98.9%)	0.01
	True	88 (1.22%)	59 (1.1%)	
disk	False	6,352 (87.73%)	4,554 (85.2%)	0.076
	True	888 (12.27%)	788 (14.8%)	
finisher	False	5,978 (82.57%)	4,437 (83.1%)	0.013
	True	1,262 (17.43%)	905 (16.9%)	
moldboard	False	7,228 (99.83%)	5,342 (100%)	0.041
	True	12 (0.17%)	0 (0%)	
ridge_till	False	7,240 (100.00%)	5,342 (100%)	< 0.001
row_cultivator	False	7,132 (98.51%)	5,342 (100%)	0.123
	True	108 (1.49%)	0 (0%)	
strip_till	False	6,536 (90.28%)	4,889 (91.5%)	0.042
	True	704 (9.72%)	453 (8.5%)	
vertical_till	False	5,877 (81.17%)	4,583 (85.8%)	0.118
	True	1,363 (18.83%)	759 (14.2%)	
lrr_code	M	4,868 (67.24%)	3,574 (66.9%)	0.137
	H	1,294 (17.87%)	1,085 (20.3%)	
	O	444 (6.13%)	216 (4.0%)	

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Table 7: Pre-Stratum E

		All (7,240)	Observed (5,342)	SMD
	F	371 (5.12%)	305 (5.7%)	
	N	139 (1.92%)	112 (2.1%)	
	P	48 (0.66%)	9 (0.2%)	
	T	33 (0.46%)	22 (0.4%)	
	L	24 (0.33%)	0 (0%)	
	K	19 (0.26%)	19 (0.4%)	

Table 8: Pre-Stratum F

		All (9,111)	Observed (5,479)	SMD
bulk_density_g_per_cm3		1.35 (0.13)	1.36 (0.14)	0.023
average_tmax_c		19.32 (4.15)	19.53 (4.04)	0.05
clay_pct		28.70 (13.37)	28.04 (12.69)	-0.049
sand_pct		22.50 (18.67)	24.15 (19.89)	0.089
elevation_m		336.89 (232.12)	318.09 (218.72)	-0.081
silt_pct		48.80 (15.88)	47.80 (16.13)	-0.063
average_precip_cm		0.24 (0.08)	0.25 (0.08)	0.089
percentage_residue_removed		0.38 (4.12)	0.10 (2.24)	-0.069
tillage_depth_cm		4.00 (4.82)	4.43 (5.12)	0.088
average_srad_mj_m2		0.67 (0.05)	0.67 (0.05)	0.052
soil_ph		6.48 (0.81)	6.45 (0.87)	-0.047
average_relative_humidity_perc		66.79 (6.53)	67.50 (5.78)	0.109
soil_carbon_pct		1.69 (2.06)	1.75 (2.54)	0.031
continuous_deep_till_to_no_till	False	8,919 (97.89%)	5,479 (100%)	0.147
	True	192 (2.11%)	0 (0%)	
cover_crop_introduction	False	8,247 (90.52%)	4,838 (88.3%)	0.076

Continued on next page

Table 8: Pre-Stratum F

		All (9,111)	Observed (5,479)	SMD
fertilizer_application_delayed	True	864 (9.48%)	641 (11.7%)	
	False	8,679 (95.26%)	5,200 (94.9%)	0.016
fertilizer_method_impact_reduction	True	432 (4.74%)	279 (5.1%)	
	False	9,001 (98.79%)	5,436 (99.2%)	0.039
legume_introduced_in_cover_crop_mix	True	110 (1.21%)	43 (0.8%)	
	False	9,061 (99.45%)	5,455 (99.6%)	0.015
longer_duration_cover_crop	True	50 (0.55%)	24 (0.4%)	
	False	8,870 (97.35%)	5,347 (97.6%)	0.015
new_cash_crops_in_rotation	True	241 (2.65%)	132 (2.4%)	
	False	8,030 (88.14%)	4,745 (86.6%)	0.047
nitrogen_stabilizer_added	True	1,081 (11.86%)	734 (13.4%)	
	False	9,027 (99.08%)	5,467 (99.8%)	0.074
no_till_introduced	True	84 (0.92%)	12 (0.2%)	
	True	4,977 (54.63%)	2,750 (50.2%)	0.089
tillage_impact_reduction	False	4,134 (45.37%)	2,729 (49.8%)	
	True	7,935 (87.09%)	4,652 (84.9%)	0.065
tillage_passes_reduction	False	1,176 (12.91%)	827 (15.1%)	
	True	7,530 (82.65%)	4,368 (79.7%)	0.077
corn	False	1,581 (17.35%)	1,111 (20.3%)	
	False	4,742 (52.05%)	2,965 (54.1%)	0.041
cotton	True	4,369 (47.95%)	2,514 (45.9%)	
	False	8,780 (96.37%)	5,173 (94.4%)	0.104
soybean	True	331 (3.63%)	306 (5.6%)	
	False	4,704 (51.63%)	2,963 (54.1%)	0.049
wheat	True	4,407 (48.37%)	2,516 (45.9%)	
	False	6,184 (67.87%)	3,918 (71.5%)	0.078
fallow	True	2,927 (32.13%)	1,561 (28.5%)	
	False	8,673 (95.19%)	5,352 (97.7%)	0.116

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Table 8: Pre-Stratum F

		All (9,111)	Observed (5,479)	SMD
chisel	True	438 (4.81%)	127 (2.3%)	
	False	8,823 (96.84%)	5,312 (97.0%)	0.006
deep_ripper	True	288 (3.16%)	167 (3.0%)	
	False	8,933 (98.05%)	5,340 (97.5%)	0.042
disk	True	178 (1.95%)	139 (2.5%)	
	False	8,049 (88.34%)	4,649 (84.9%)	0.109
finisher	True	1,062 (11.66%)	830 (15.1%)	
	False	7,787 (85.47%)	4,719 (86.1%)	0.019
moldboard	True	1,324 (14.53%)	760 (13.9%)	
	False	9,111 (100.00%)	5,479 (100%)	< 0.001
ridge_till	False	9,027 (99.08%)	5,410 (98.7%)	0.035
	True	84 (0.92%)	69 (1.3%)	
row_cultivator	False	8,967 (98.42%)	5,396 (98.5%)	0.005
	True	144 (1.58%)	83 (1.5%)	
strip_till	False	8,313 (91.24%)	4,871 (88.9%)	0.083
	True	798 (8.76%)	608 (11.1%)	
vertical_till	False	7,698 (84.49%)	4,931 (90.0%)	0.152
	True	1,413 (15.51%)	548 (10.0%)	
lrr_code	M	3,391 (37.22%)	2,315 (42.3%)	0.256
	P	1,400 (15.37%)	1,046 (19.1%)	
	F	1,082 (11.88%)	479 (8.7%)	
	G	930 (10.21%)	582 (10.6%)	
	O	830 (9.11%)	523 (9.5%)	
	H	774 (8.50%)	323 (5.9%)	
	K	370 (4.06%)	45 (0.8%)	
	L	167 (1.83%)	118 (2.2%)	
	S	60 (0.66%)	11 (0.2%)	
	R	60 (0.66%)	24 (0.4%)	

Continued on next page

Table 8: Pre-Stratum F

		All (9,111)	Observed (5,479)	SMD
	J	24 (0.26%)	0 (0%)	
	N	23 (0.25%)	13 (0.2%)	

Table 9: Pre-Stratum G

		All (1,449)	Observed (1,190)	SMD
bulk_density_g_per_cm3		1.36 (0.11)	1.35 (0.11)	-0.114
average_tmax_c		19.37 (3.18)	19.13 (2.89)	-0.075
clay_pct		25.36 (9.05)	25.11 (6.82)	-0.028
sand_pct		22.55 (18.54)	21.46 (17.04)	-0.059
elevation_m		398.00 (257.04)	421.91 (248.05)	0.093
silt_pct		52.08 (16.36)	53.43 (15.90)	0.082
average_precip_cm		0.22 (0.08)	0.21 (0.07)	-0.101
percentage_residue_removed		0.51 (4.34)	0.45 (4.50)	-0.013
tillage_depth_cm		3.18 (4.09)	3.11 (4.07)	-0.019
average_srad_mj_m2		0.67 (0.05)	0.67 (0.05)	-0.048
soil_ph		6.46 (0.74)	6.52 (0.71)	0.086
average_relative_humidity_perc		65.22 (7.75)	64.65 (7.65)	-0.075
soil_carbon_pct		1.60 (1.01)	1.66 (1.04)	0.067
continuous_deep_till_to_no_till	False	1,449 (100.00%)	1,190 (100%)	< 0.001
cover_crop_introduction	False	1,238 (85.44%)	1,024 (86.1%)	0.017
	True	211 (14.56%)	166 (13.9%)	
fertilizer_application_delayed	False	1,204 (83.09%)	984 (82.7%)	0.011
	True	245 (16.91%)	206 (17.3%)	
fertilizer_method_impact_reduction	False	1,425 (98.34%)	1,166 (98.0%)	0.028
	True	24 (1.66%)	24 (2.0%)	

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Table 9: Pre-Stratum G

		All (1,449)	Observed (1,190)	SMD
legume_introduced_in_cover_crop_mix	False	1,425 (98.34%)	1,190 (100%)	0.13
	True	24 (1.66%)	0 (0%)	
longer_duration_cover_crop	False	1,413 (97.52%)	1,178 (99.0%)	0.095
	True	36 (2.48%)	12 (1.0%)	
new_cash_crops_in_rotation	False	1,342 (92.62%)	1,126 (94.6%)	0.077
	True	107 (7.38%)	64 (5.4%)	
nitrogen_stabilizer_added	False	1,400 (96.62%)	1,159 (97.4%)	0.043
	True	49 (3.38%)	31 (2.6%)	
no_till_introduced	False	1,071 (73.91%)	857 (72.0%)	0.043
	True	378 (26.09%)	333 (28.0%)	
tillage_impact_reduction	False	744 (51.35%)	590 (49.6%)	0.035
	True	705 (48.65%)	600 (50.4%)	
tillage_passes_reduction	False	794 (54.80%)	640 (53.8%)	0.02
	True	655 (45.20%)	550 (46.2%)	
corn	False	729 (50.31%)	584 (49.1%)	0.025
	True	720 (49.69%)	606 (50.9%)	
cotton	False	1,401 (96.69%)	1,156 (97.1%)	0.025
	True	48 (3.31%)	34 (2.9%)	
soybean	False	862 (59.49%)	733 (61.6%)	0.043
	True	587 (40.51%)	457 (38.4%)	
wheat	False	1,060 (73.15%)	909 (76.4%)	0.073
	True	389 (26.85%)	281 (23.6%)	
fallow	False	1,427 (98.48%)	1,170 (98.3%)	0.013
	True	22 (1.52%)	20 (1.7%)	
chisel	False	1,401 (96.69%)	1,144 (96.1%)	0.031
	True	48 (3.31%)	46 (3.9%)	
deep_ripper	False	1,389 (95.86%)	1,142 (96.0%)	0.005
	True	60 (4.14%)	48 (4.0%)	

Continued on next page

Table 9: Pre-Stratum G

		All (1,449)	Observed (1,190)	SMD
disk	False	1,368 (94.41%)	1,133 (95.2%)	0.035
	True	81 (5.59%)	57 (4.8%)	
finisher	False	1,115 (76.95%)	917 (77.1%)	0.003
	True	334 (23.05%)	273 (22.9%)	
moldboard	False	1,449 (100.00%)	1,190 (100%)	< 0.001
ridge_till	False	1,449 (100.00%)	1,190 (100%)	< 0.001
row_cultivator	False	1,449 (100.00%)	1,190 (100%)	< 0.001
strip_till	False	1,368 (94.41%)	1,133 (95.2%)	0.035
	True	81 (5.59%)	57 (4.8%)	
vertical_till	False	1,248 (86.13%)	1,014 (85.2%)	0.027
	True	201 (13.87%)	176 (14.8%)	
lrr_code	M	664 (45.82%)	590 (49.6%)	NaN
	H	427 (29.47%)	370 (31.1%)	
	P	130 (8.97%)	101 (8.5%)	
	F	120 (8.28%)	94 (7.9%)	
	O	36 (2.48%)	0 (0%)	
	N	36 (2.48%)	23 (1.9%)	
	S	12 (0.83%)	0 (0%)	
	K	12 (0.83%)	12 (1.0%)	
	T	12 (0.83%)	0 (0%)	

Table 10: Pre-Stratum H

	All (2,994)	Observed (0)	SMD
bulk_density_g_per_cm3	1.39 (0.10)	nan (nan)	NaN
average_tmax_c	21.74 (3.57)	nan (nan)	NaN

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Table 10: Pre-Stratum H

		All (2,994)	Observed (0)	SMD
clay_pct		26.25 (12.29)	nan (nan)	NaN
sand_pct		25.00 (22.56)	nan (nan)	NaN
elevation_m		283.50 (256.18)	nan (nan)	NaN
silt_pct		48.75 (19.15)	nan (nan)	NaN
average_precip_cm		0.27 (0.09)	nan (nan)	NaN
percentage_residue_removed		0.49 (4.58)	nan (nan)	NaN
tillage_depth_cm		3.35 (3.82)	nan (nan)	NaN
average_srad_mj_m2		0.70 (0.05)	nan (nan)	NaN
soil_ph		6.11 (0.88)	nan (nan)	NaN
average_relative_humidity_perc		67.09 (7.52)	nan (nan)	NaN
soil_carbon_pct		1.18 (0.83)	nan (nan)	NaN
continuous_deep_till_to_no_till	False	2,994 (100.00%)	0 (0%)	< 0.001
cover_crop_introduction	False	2,190 (73.15%)	0 (0%)	NaN
	True	804 (26.85%)	0 (0%)	
fertilizer_application_delayed	False	2,238 (74.75%)	0 (0%)	NaN
	True	756 (25.25%)	0 (0%)	
fertilizer_method_impact_reduction	False	2,898 (96.79%)	0 (0%)	NaN
	True	96 (3.21%)	0 (0%)	
legume_introduced_in_cover_crop_mix	False	2,958 (98.80%)	0 (0%)	NaN
	True	36 (1.20%)	0 (0%)	
longer_duration_cover_crop	False	2,862 (95.59%)	0 (0%)	NaN
	True	132 (4.41%)	0 (0%)	
new_cash_crops_in_rotation	False	2,576 (86.04%)	0 (0%)	NaN
	True	418 (13.96%)	0 (0%)	
nitrogen_stabilizer_added	False	2,874 (95.99%)	0 (0%)	NaN
	True	120 (4.01%)	0 (0%)	
no_till_introduced	False	1,950 (65.13%)	0 (0%)	NaN
	True	1,044 (34.87%)	0 (0%)	

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Table 10: Pre-Stratum H

		All (2,994)	Observed (0)	SMD
tillage_impact_reduction	True	2,160 (72.14%)	0 (0%)	NaN
	False	834 (27.86%)	0 (0%)	
tillage_passes_reduction	True	2,028 (67.74%)	0 (0%)	NaN
	False	966 (32.26%)	0 (0%)	
corn	True	1,678 (56.05%)	0 (0%)	NaN
	False	1,316 (43.95%)	0 (0%)	
cotton	False	2,766 (92.38%)	0 (0%)	NaN
	True	228 (7.62%)	0 (0%)	
soybean	True	1,868 (62.39%)	0 (0%)	NaN
	False	1,126 (37.61%)	0 (0%)	
wheat	False	1,818 (60.72%)	0 (0%)	NaN
	True	1,176 (39.28%)	0 (0%)	
fallow	False	2,970 (99.20%)	0 (0%)	NaN
	True	24 (0.80%)	0 (0%)	
chisel	False	2,934 (98.00%)	0 (0%)	NaN
	True	60 (2.00%)	0 (0%)	
deep_ripper	False	2,934 (98.00%)	0 (0%)	NaN
	True	60 (2.00%)	0 (0%)	
disk	False	2,806 (93.72%)	0 (0%)	NaN
	True	188 (6.28%)	0 (0%)	
finisher	False	2,050 (68.47%)	0 (0%)	NaN
	True	944 (31.53%)	0 (0%)	
moldboard	False	2,994 (100.00%)	0 (0%)	< 0.001
ridge_till	False	2,994 (100.00%)	0 (0%)	< 0.001
row_cultivator	False	2,982 (99.60%)	0 (0%)	NaN
	True	12 (0.40%)	0 (0%)	
strip_till	False	2,754 (91.98%)	0 (0%)	NaN
	True	240 (8.02%)	0 (0%)	

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Table 10: Pre-Stratum H

		All (2,994)	Observed (0)	SMD
vertical_till	False	2,778 (92.79%)	0 (0%)	NaN
	True	216 (7.21%)	0 (0%)	
lrr_code	M	812 (27.12%)	0 (0%)	NaN
	P	756 (25.25%)	0 (0%)	
	H	610 (20.37%)	0 (0%)	
	O	468 (15.63%)	0 (0%)	
	N	84 (2.81%)	0 (0%)	
	K	72 (2.40%)	0 (0%)	
	L	60 (2.00%)	0 (0%)	
	F	36 (1.20%)	0 (0%)	
	J	36 (1.20%)	0 (0%)	
	T	36 (1.20%)	0 (0%)	
	I	24 (0.80%)	0 (0%)	

Figures 2–9 show the absolute SMDs from Tables 3–10.

5 Discussion and Conclusion

In the majority of cases in which reversible emissions reduction could not be estimated, the cause was logistical and is unlikely to be related to the soil attributes being measured (%SOC, bulk density, pH, and texture). In particular, nearly all missingness occurred prior to model runs (see Table 1), and these missing points are mostly due to sampling logistics (see Table 2).

The SMDs in Tables 3–10 and Figures 2–9 are nearly all less than the 0.1 threshold commonly used in other domains, and all are less than the 0.25 threshold that is sometimes used instead of 0.1 (see Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) for background on the 0.1 and 0.25 thresholds). The exception is stratum H, which has been randomized, but no samples have been received as of this submission. As a result, the SMDs for stratum H are not yet defined, but are expected to be defined and small in the second submission.

While the 0.1 and 0.25 thresholds suggested in Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) are useful starting points for Soil Enrichment Protocol (SEP) projects, it is not clear whether these are the most appropriate thresholds to use for SEP projects. As noted in Supporting Document `IndigoCarbon_US-1_2023_0062` (2023), experience specific to SEP projects, such as the analyses in the present document, will help to determine appropriate thresholds, as well as which attributes are most important for inclusion in Tables 3–10 and Figures 2–9.

Given that: (1) the majority of missing points are missing for reasons that are unlikely to cause bias, and (2) the absolute SMDs are mostly less than 0.25, we have concluded that there is minimal concern for bias due to missing data. Therefore, we have not post-stratified and are only submitting an unadjusted analysis.

In addition to the tabulation of reasons for missingness in Table 2, Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) requests data on: (1) the date at which the problem occurred or was reported, and (2) a 1-2 sentence description of the problem. In compiling this report, we thought that this additional information would not be helpful, and we plan to request a revision to Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) to remove these requirements for future reporting periods. However, if helpful, we could provide dates for all missing soil samples except those for which a field staff has not yet been assigned, or for which a field staff has been assigned but has not yet visited the field. We could also provide slightly more detailed reasons for some but not all missing soil samples. (As Indigo's technology developed, more tools were created to record the reason why a sample was not taken, so older sampling has sparser information about reasons for missingness.)

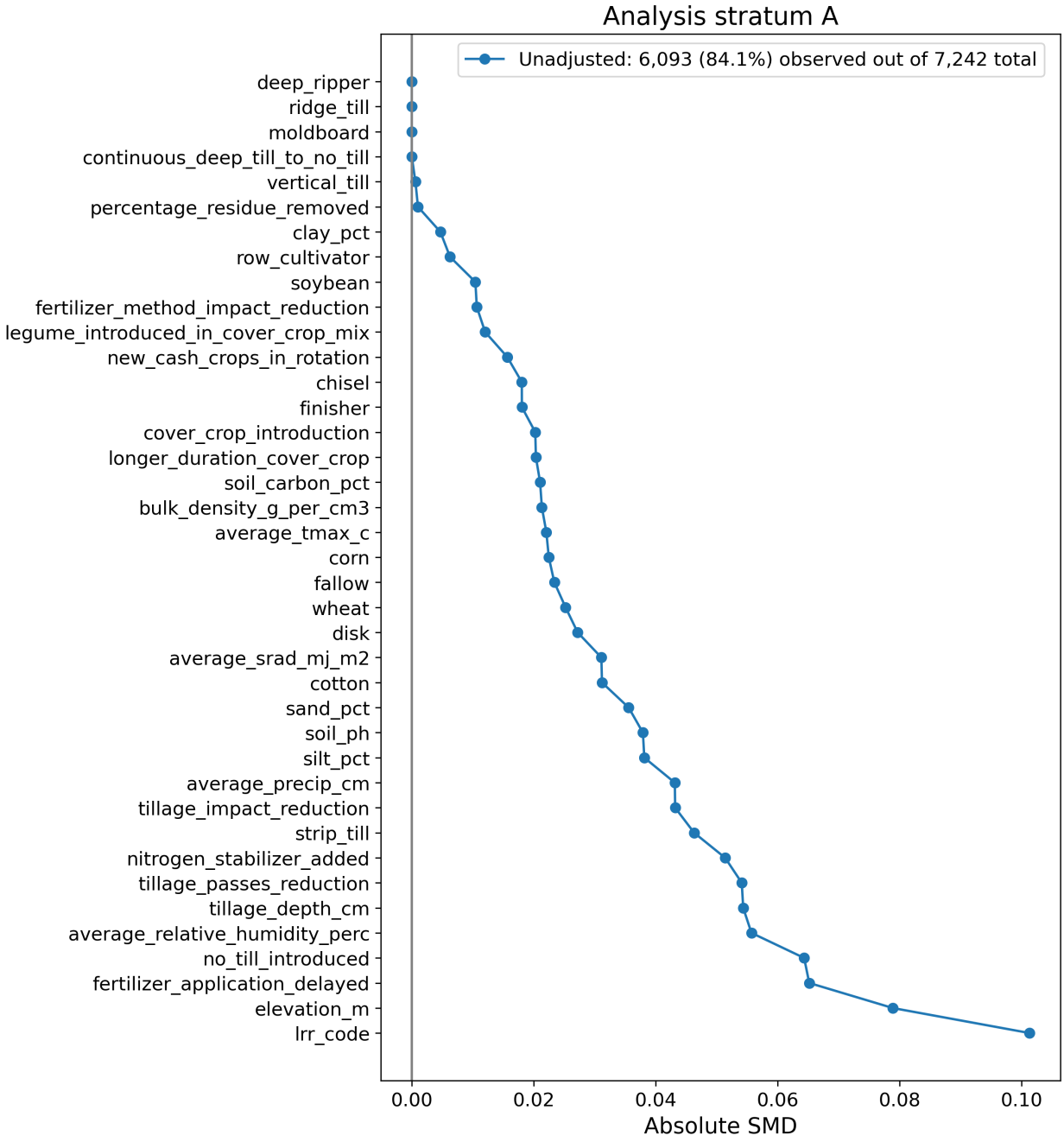


Figure 2: Absolute standardized mean differences (SMDs) for Analysis Stratum A

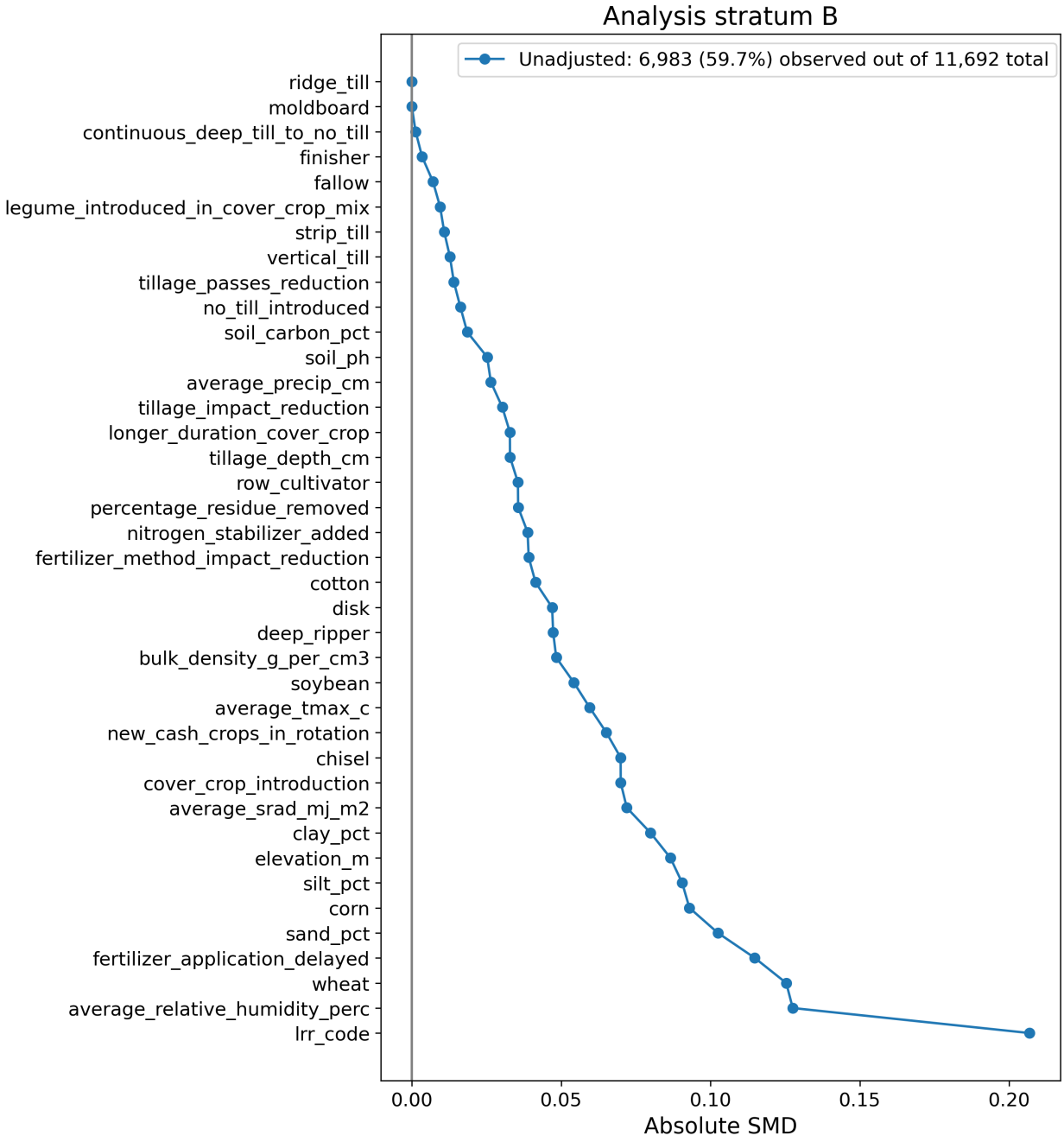


Figure 3: Absolute standardized mean differences (SMDs) for Analysis Stratum B

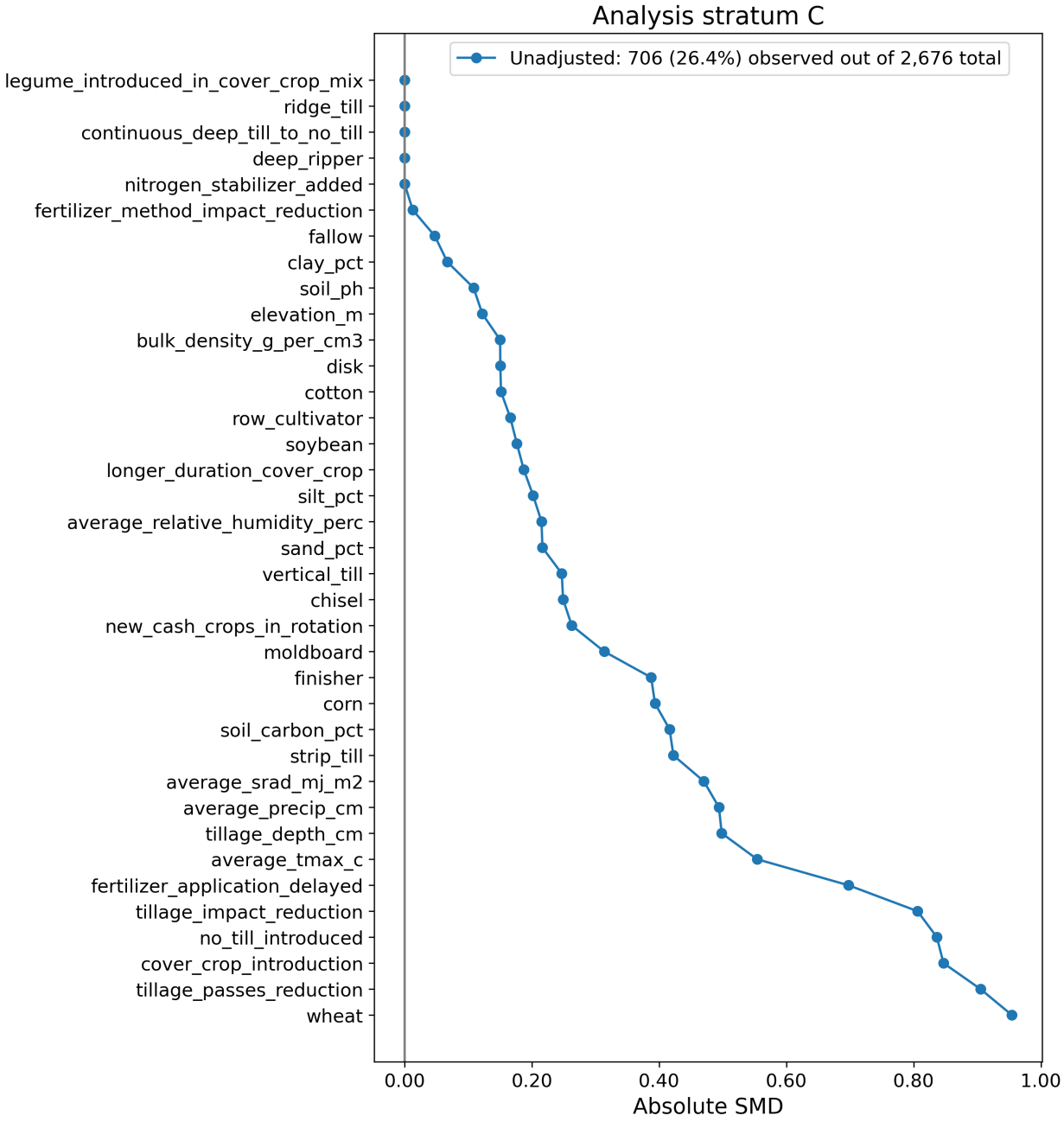


Figure 4: Absolute standardized mean differences (SMDs) for Analysis Stratum C

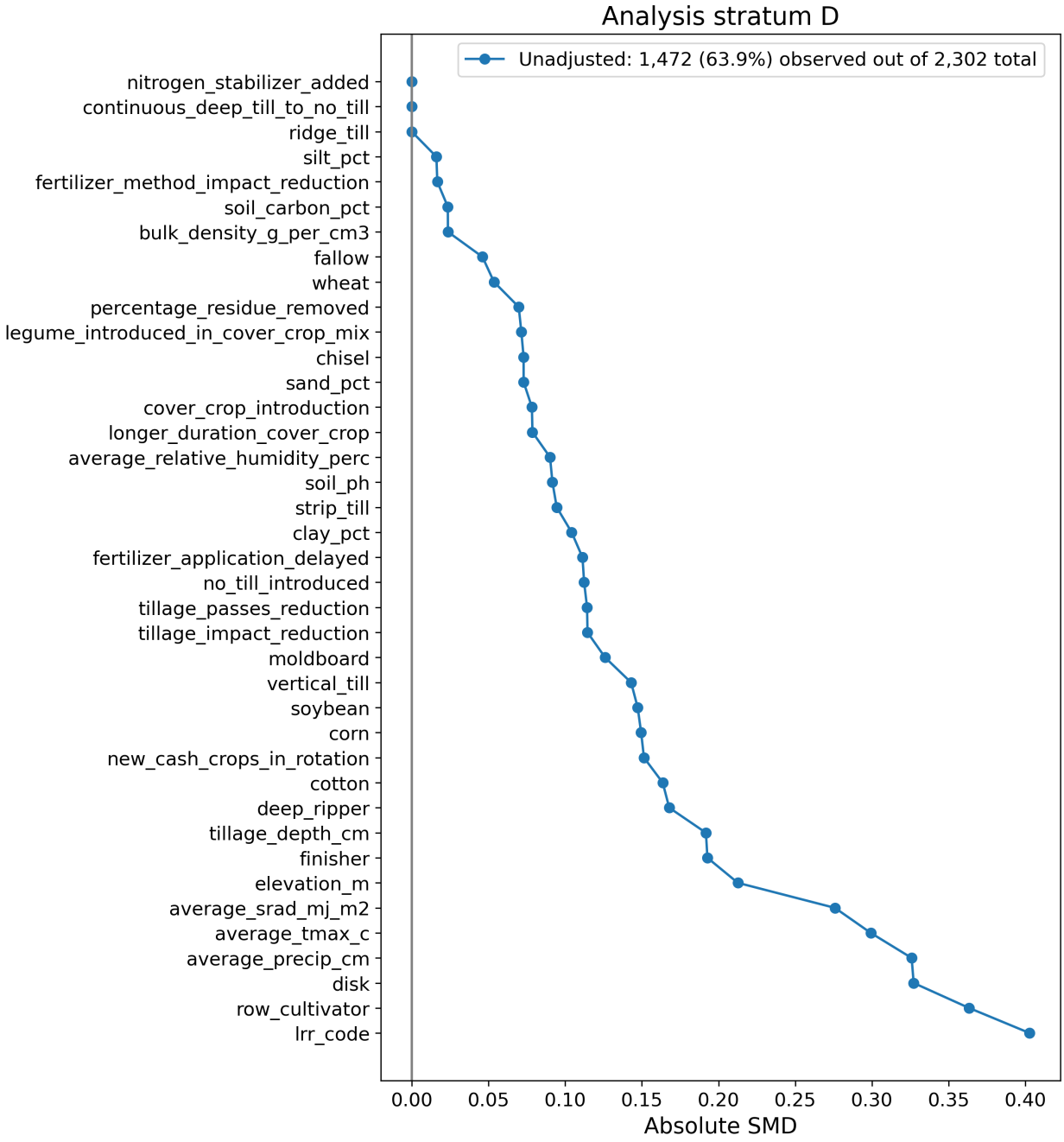


Figure 5: Absolute standardized mean differences (SMDs) for Analysis Stratum D

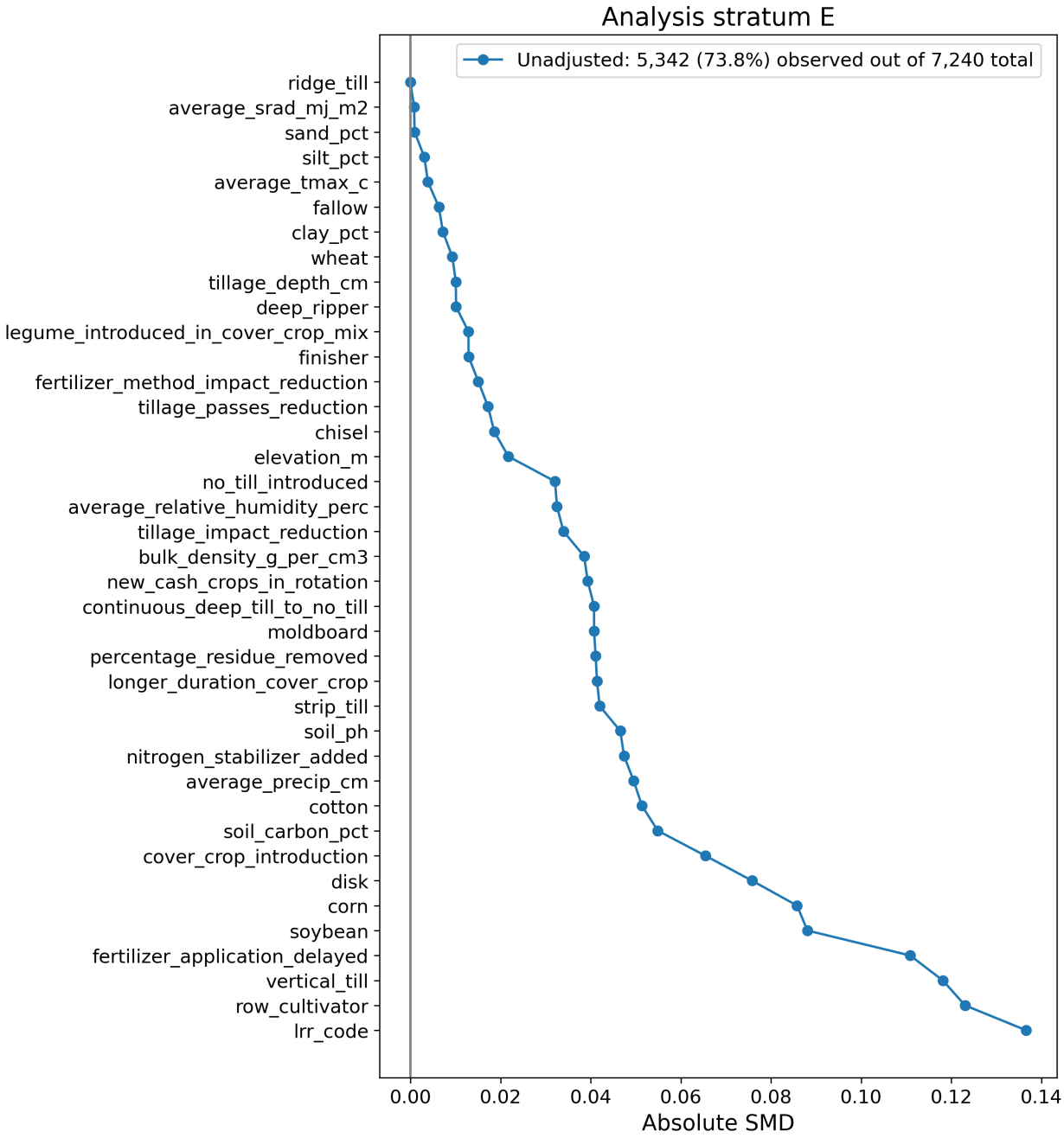


Figure 6: Absolute standardized mean differences (SMDs) for Analysis Stratum E

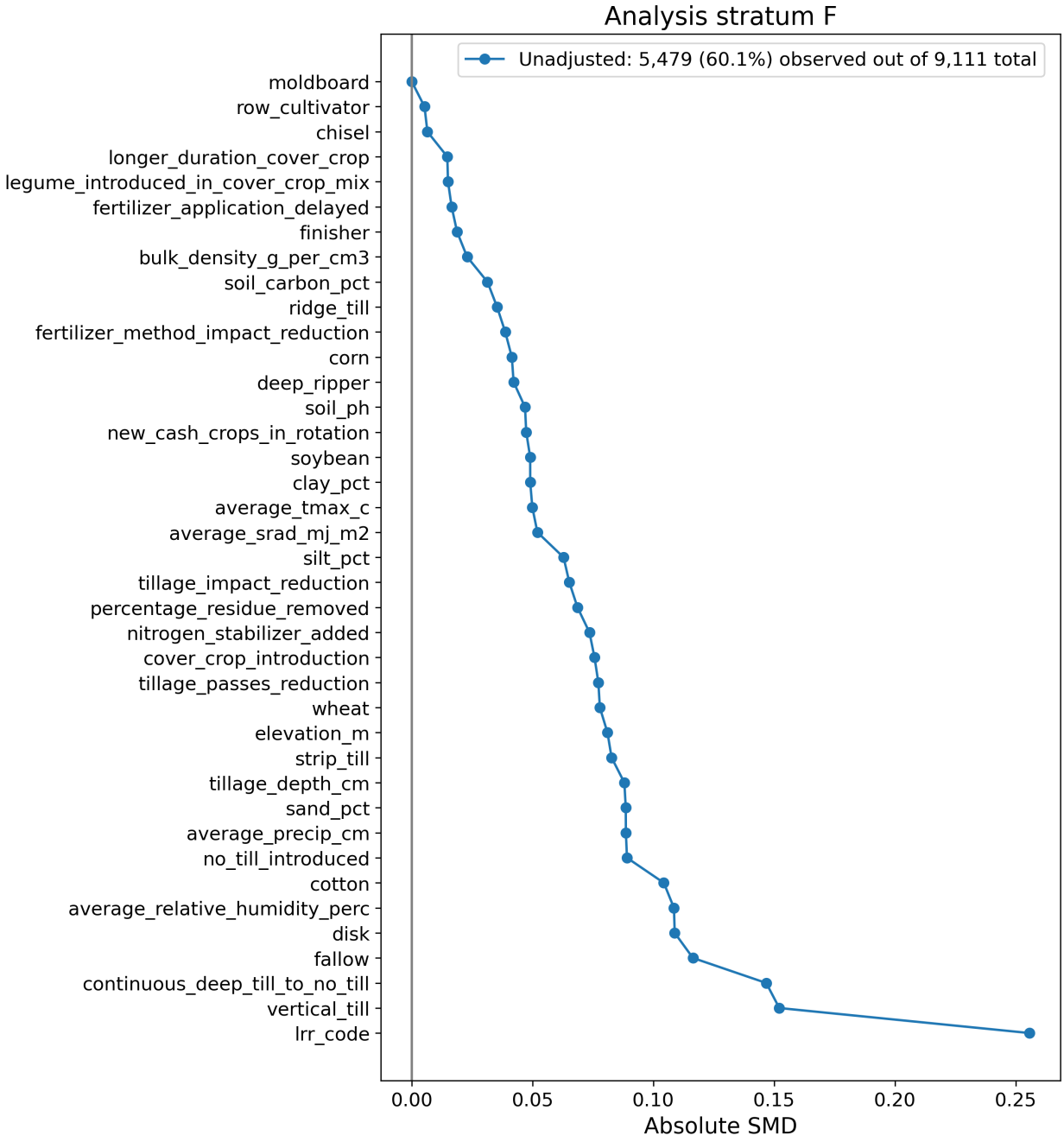


Figure 7: Absolute standardized mean differences (SMDs) for Analysis Stratum F

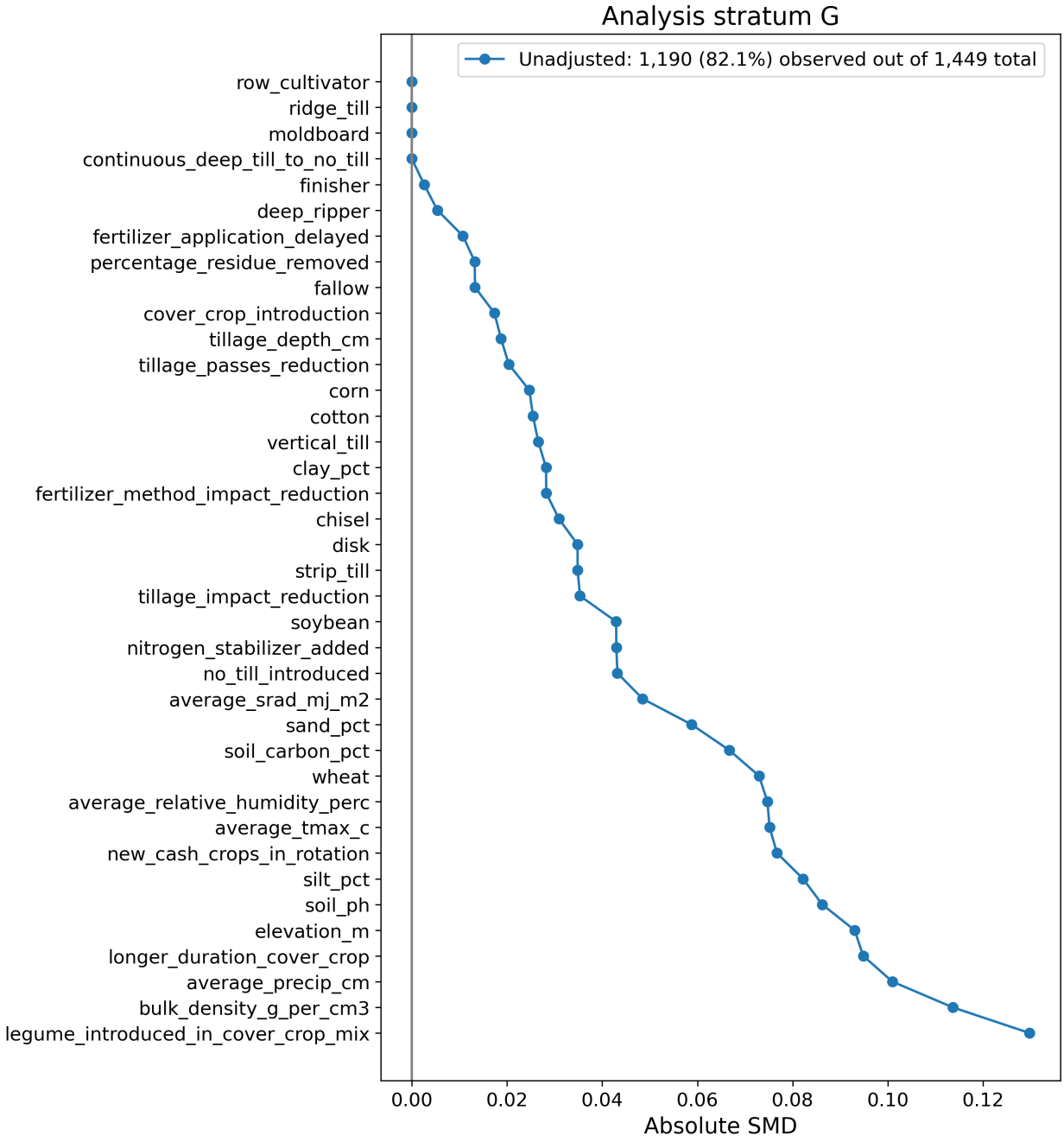


Figure 8: Absolute standardized mean differences (SMDs) for Analysis Stratum G

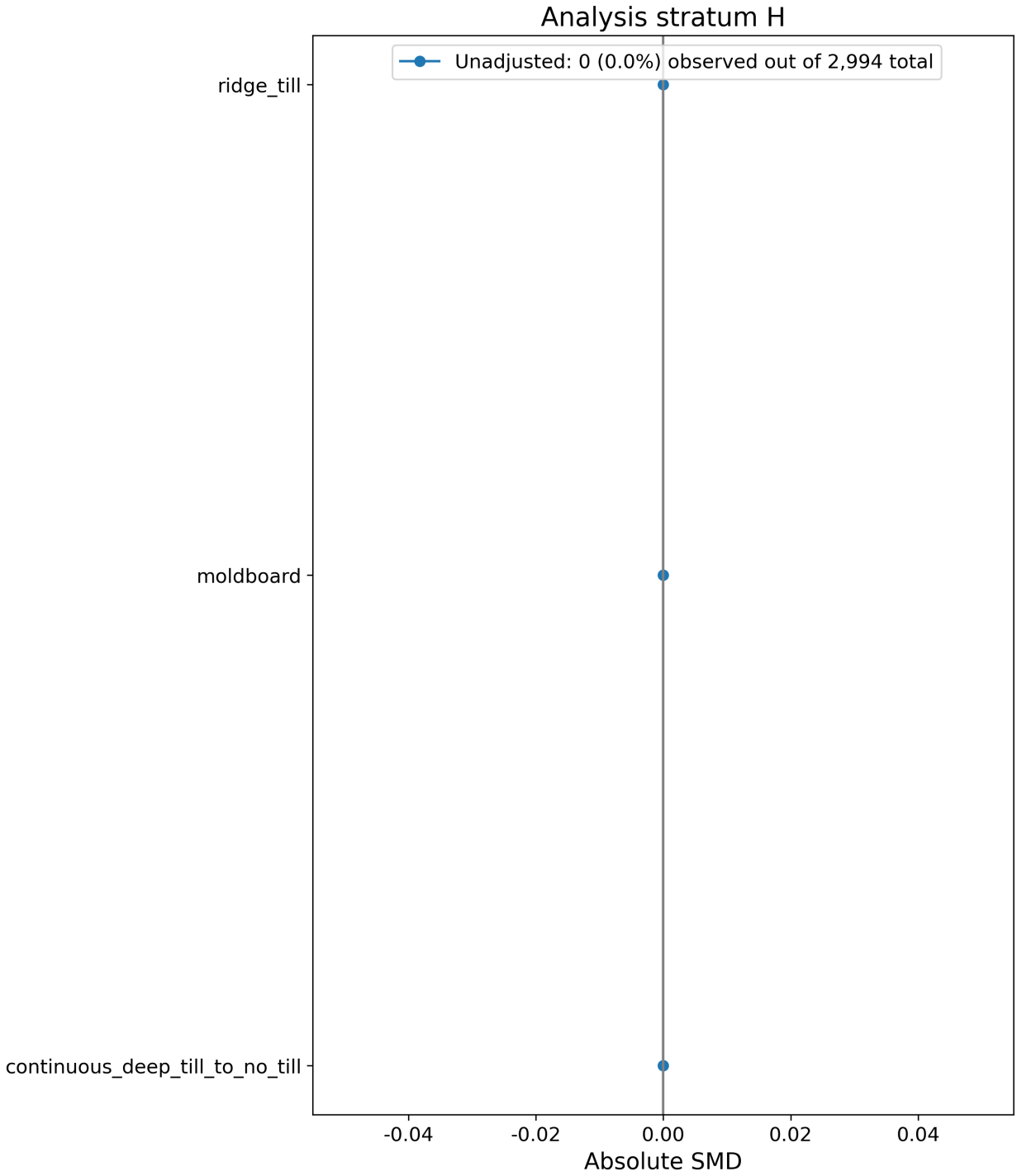


Figure 9: Absolute standardized mean differences (SMDs) for Analysis Stratum H

A Attribute Definitions

Table 11 shows the definitions and source for each of the attributes shown in Tables 3–10 and Figures 2–9.

Table 11: Attribute definitions

Attribute	Definition	Source
lrr_code	Land resource region	NRCS
soil_carbon_pct	Soil carbon percent (units: percent); temporal mean	SSURGO
bulk_density_g_per_cm3	Bulk density (units: grams / cubic cm); temporal mean	SSURGO
sand_pct	Sand (units: percent); temporal mean	SSURGO
silt_pct	Silt (units: percent); temporal mean	SSURGO
clay_pct	Clay (units: percent); temporal mean	SSURGO
soil_ph	pH; temporal mean	SSURGO
elevation_m	Elevation (units: meters); temporal mean	NLDAS
average_tmin	Daily minimum temperature (units: deg C); temporal mean	NLDAS
average_tmax_c	Daily maximum temperature (units: deg C); temporal mean	NLDAS
average_precip_cm	Daily precipitation (units: cm); temporal mean	NLDAS
average_relative_humidity_perc	Daily average relative humidity (units: percent); temporal mean	NLDAS
average_srad_mj_m2	Daily average solar radiation (units: MJ / m^2); temporal mean	NLDAS
cover_crop_introduction	True if this practice change was adopted, and false otherwise	Grower
fertilizer_application_delayed	True if this practice change was adopted, and false otherwise	Grower
fertilizer_form_impact_reduction	True if this practice change was adopted, and false otherwise	Grower
fertilizer_method_impact_reduction	True if this practice change was adopted, and false otherwise	Grower
fertilizer_reduction	True if this practice change was adopted, and false otherwise	Grower
longer_duration_cover_crop	True if this practice change was adopted, and false otherwise	Grower
new_cash_crops_in_rotation	True if this practice change was adopted, and false otherwise	Grower
nitrogen_stabilizer_added	True if this practice change was adopted, and false otherwise	Grower
no-till_introduced	True if this practice change was adopted, and false otherwise	Grower
synthetic_to_organic_n	True if this practice change was adopted, and false otherwise	Grower
tillage_delayed	True if this practice change was adopted, and false otherwise	Grower
tillage_impact_reduction	True if this practice change was adopted, and false otherwise	Grower

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Table 11: Attribute definitions

Attribute	Definition	Source
tillage_passes_reduction	True if this practice change was adopted, and false otherwise	Grower
wheat	True if a planted crop falls into the wheat crop functional (see Table 12)	Grower
soybean	True if a planted crop falls into the soybean crop functional (see Table 12)	Grower
corn	True if a planted crop falls into the corn crop functional (see Table 12)	Grower
fallow	True if no crop was planted in that cultivation cycle	Grower
percentage_residue_removed	Percent of residual removed (units: percent)	Grower
till_method	The tillage method	Grower
depth_cm	The tillage depth	Grower

Table 12: Names of crop functional groups

	Shorthand name	Name used in database
0	wheat	wheat-barley-rye-oats-flax-canola-potato-tobaco-sunflower-safflower-sugarbeat-cool-grass-triticale
1	soybean	soybeans-dry-beans-peas-peanuts-alfafa-clover
2	corn	corn-millet-sorghum-sudangrass
3	cotton	cotton

Changes in this supporting document since the last verification

- New strata included
- Discussion of results updated

References

[1] Supporting document `IndigoCarbon_US-1_2023_0001`. *Soil Carbon (30 cm) Sampling*. 2024.

[2] Supporting document `IndigoCarbon_US-1_2023_0062`. *Addressing incomplete soil sample data in Soil Enrichment Protocol projects (v4.4.1)*. 2023.

[3] Supporting document `IndigoCarbon_US-1_2023_0065`. *Model output postprocessing*. 2023.

[4] Supporting document `IndigoCarbon_US-1_2024_0034`. *Boundary review workflow and standard operating procedures*. 2023.

[5] Supporting document `IndigoCarbon_US-1_2024_0068`. *Details on the pre-strata*. 2024.